

MAT 534 HW01

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Problem 1. Suppose that G is a set with an associative operation, written as multiplication. Show that G is a group if and only if, for every a and b in G , the equations $xa = b$ and $ay = b$ have solutions $x, y \in G$.

Solution. If G is a group, then $xa = b$ has the solution $x = ba^{-1}$ and $ay = b$ has the solution $y = a^{-1}b$.

Conversely suppose that $xa = b$ and $ay = b$ have solutions for all $a, b \in G$. Fixing $a \in G$, let e be an element of G such that $ea = a$. Given $b \in G$, pick y so that $ay = b$. Then,

$$eb = e(ay) = (ea)y = ay = b$$

so multiplication on the left by e fixes all elements of G . Similarly, there exists an $e' \in G$ such that $ae' = a$, and then, given an arbitrary b in G , if $xa = b$, we get

$$be' = (xa)e' = x(ae') = xa = b$$

so multiplication on the right by e' fixes all elements of G . This implies that

$$e = ee' = e'$$

so in fact e is the identity element for the operation on G .

It remains to check the existence of inverses in G . For some $a \in G$, we are given that there exists a left inverse x (so $xa = e$) and a right inverse y (so $ay = e$); we need to check that $x = y$. This follows from

$$x = x(ay) = (xa)y = y,$$

so for any $a \in G$ there exists x such that $ax = xa = e$. □

Problem 2. Suppose G is a group in which $x^2 = 1$ for every $x \in G$. Show that G is abelian.

Solution. Since $x^2 = 1$ for all x , each element of G is its own inverse. This means that for any $a, b \in G$,

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba$$

so the operation of G is commutative. □

Problem 3. Let G be a group and o be any element of G . Define a new operation $*$ on G by the formula $x * y = xoy$.

- (a) Show that G is a group under the new operation $*$. What is the identity of G for the new operation? If $x \in G$, what is the inverse of x for the new operation?
- (b) If G is a group and $o \in G$, let G_o denote the group defined in (a). Show that $G \cong G_o$.

Solution. (a) First, $*$ is associative because

$$(x * y) * z = (xoy)oz = xo(yoz) = x * (y * z),$$

using the fact that the operation of G is of course associative. Second, the identity element for the operation $*$ is o^{-1} , since for any $x \in G$,

$$x * o^{-1} = xoo^{-1} = x, \quad o^{-1} * x = o^{-1}ox = x.$$

Finally, given $x \in G$, the inverse of x for $*$ is $(oxo)^{-1} = o^{-1}x^{-1}o^{-1}$, as it is easy to see that

$$xo(oxo)^{-1} = (oxo)^{-1}xo = o^{-1}.$$

Thus $(G, *)$ satisfies the axioms of a group.

(b) Define $\varphi: G \rightarrow G_o$ by $\varphi(x) = o^{-1}x$ (where the inverse o^{-1} is with respect to the operation of G). This is a bijective function because it is invertible via the map $y \mapsto oy$, and it is a homomorphism because

$$\varphi(xy) = o^{-1}(xy) = (o^{-1}x)o(o^{-1}y) = \varphi(x) * \varphi(y),$$

so φ is an isomorphism $G \rightarrow G_o$. □

Problem 4. (a) Let G be the set of linear polynomials $f(x) = ax + b$, where $a, b \in \mathbb{R}$ and $a \neq 0$. Show that G is a group under composition of functions.

(b) Let

$$H = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a, b \in \mathbb{R}, a \neq 0 \right\}.$$

Show that H is a group under matrix multiplication.

(c) Show that G and H are isomorphic.

Solution. (a) Composing two linear polynomials gives another linear polynomial: if $f(x) = ax + b$ and $g(x) = cx + d$, then

$$(g \circ f)(x) = c(ax + b) + d = acx + (bc + d),$$

so G is closed under composition. Function composition is of course associative, with identity being the identity function $e(x) = x$ for all x . Finally, the inverse of $f(x) = ax + b$ is

$$f^{-1}(x) = \frac{x}{a} - \frac{b}{a},$$

since

$$\begin{aligned} (f^{-1} \circ f)(x) &= \frac{ax + b}{a} - \frac{b}{a} = x + \frac{b}{a} - \frac{b}{a} = x \\ (f \circ f^{-1})(x) &= a \left(\frac{x}{a} - \frac{b}{a} \right) + b = x - b + b = x. \end{aligned}$$

(b) To see that H is closed under matrix multiplication, we compute

$$\begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ac & bc + d \\ 0 & 1 \end{pmatrix}.$$

Matrix multiplication is of course associative, with identity given by the identity matrix. Finally,

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1/a & -b/a \\ 0 & 1 \end{pmatrix},$$

which can be verified easily. (Less directly, we can also just note that the inverse in $GL_2(\mathbb{R})$ of an element of G has $(0 \ 1)$ as its bottom row due to how the row reduction algorithm works.)

(c) G and H are isomorphic with isomorphism $\varphi: G \rightarrow H$ given by

$$\varphi(x \mapsto ax + b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

This is clearly invertible hence bijective, as the first row of the matrix exactly encodes the coefficients of the corresponding linear polynomial in G . To check that this is a homomorphism, if $f(x) = ax + b$ and $g(x) = cx + d$ with $a, c \neq 0$, then

$$\varphi(g)\varphi(f) = \begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ac & bc + d \\ 0 & 1 \end{pmatrix} = \varphi(g \circ f),$$

recalling from part (a) that $(g \circ f)(x) = acx + (bc + d)$. □

Problem 5. Let $\mathbb{C}^\times$ be the multiplicative group of the complex numbers, i.e. the non-zero elements of $\mathbb{C}$, under multiplication. Show that $\mathbb{C}^\times$ is isomorphic to the direct product of the two additive groups $\mathbb{R}$ and $\mathbb{R}/\mathbb{Z}$.

Solution. Consider the map

$$\begin{aligned} \varphi: \mathbb{R} \times \mathbb{R} &\rightarrow \mathbb{C}^\times \\ (r, \theta) &\mapsto e^{r+2\pi i\theta} = e^r(\cos(2\pi\theta) + i\sin(2\pi\theta)). \end{aligned}$$

This is clearly a homomorphism, as

$$\varphi(r + s, \theta + \psi) = e^{r+s+2\pi i(\theta+\psi)} = e^{r+2\pi i\theta} e^{s+2\pi i\psi} = \varphi(r, \theta)\varphi(s, \psi).$$

Furthermore, the kernel of this homomorphism is the set of (r, θ) such that

$$e^{r+2\pi i\theta} = 1 \iff e^r = 1, \ 2\pi\theta \in 2\pi\mathbb{Z}$$

which is equivalent to $\theta \in \mathbb{Z}$ and $r = 0$, so

$$\mathbb{C}^\times \cong \frac{\mathbb{R} \times \mathbb{R}}{\langle (0, 1) \rangle}.$$

In turn, by the homomorphism $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \times (\mathbb{R}/\mathbb{Z})$ defined by $(a, b) \mapsto (a, b \bmod 1)$, whose kernel is $\langle (0, 1) \rangle$, we get

$$\mathbb{C}^\times \cong \frac{\mathbb{R} \times \mathbb{R}}{\langle (0, 1) \rangle} \cong \mathbb{R} \times (\mathbb{R}/\mathbb{Z}),$$

which is what we wanted to show. □

Problem 6. Let $n \geq 2$ be an integer. Determine the center of the dihedral group D_{2n} .

Solution. Recall that

$$D_{2n} = \langle r, d \mid r^n = d^2 = 1, rd = dr^{-1} \rangle = \{1, r, \dots, r^{n-1}, d, rd, \dots, r^{n-1}d\}.$$

When $n = 2$, it is easy to check that the group is abelian, hence $Z(D_{2n}) = D_{2n}$. Assume henceforth that $n > 2$.

First, note that $r^k d$ is always not in the center of D_{2n} , seeing as $r \cdot r^k d = r^k d \cdot r^{-1} \neq r^k d \cdot r$ because $r \neq r^{-1}$ (since $r^2 = 1$ would imply $n = 2$).

Thus $Z(D_{2n})$ is a subgroup of $\{1, r, \dots, r^{n-1}\}$. Obviously the powers of r all commute with each other, so we need to check which powers of r commute with the $r^k d$. Take $0 \leq m < n$. Then

$$r^k d \cdot r^m = r^m \cdot r^k d = r^k d \cdot r^{-m}$$

if and only if $r^{2m} = 1$. This requires $n \mid 2m$, which, along with the bound $0 \leq 2m < 2n$, implies $2m = 0$ or $2m = n$. The former case tells us that 1 is in the center (shocking), while the latter tells us that, if n is even, then $r^{n/2}$ is in the center. Therefore

$$Z(D_{2n}) = \begin{cases} D_4 & n = 2 \\ \{1\} & n \text{ is odd} \\ \{1, r^{n/2}\} & n \text{ is even.} \end{cases}$$

□

Problem 7. Let H and K be subgroups of G , with K normal in G .

(a) Show that $HK = \{hk : h \in H, k \in G\}$ is a subgroup of G .

(b) Suppose that H is also normal. Show that HK is a normal subgroup of G .

Solution. (a) To show that HK is a subgroup of G , we need to show that if g and g' are in HK , then so is $g(g')^{-1}$. In this spirit, let $h_1 k_1$ and $h_2 k_2$ be elements of HK , with $h_1, h_2 \in H$ and $k_1, k_2 \in K$. Then,

$$h_1 k_1 (h_2 k_2)^{-1} = h_1 k_1 k_2^{-1} h_2^{-1} = h_1 h_2^{-1} (h_2 k_1 k_2^{-1} h_2^{-1}).$$

Since K is normal in G , $k_1 k_2^{-1} \in K$ implies that the conjugate $h_2 k_1 k_2^{-1} h_2^{-1}$ is also in K , so we have expressed $h_1 k_1 (h_2 k_2)^{-1}$ as the product of an element $h_1 h_2^{-1}$ of H and an element $h_2 k_1 k_2^{-1} h_2^{-1}$ of K , hence it is in HK as desired.

(b) If H is also normal, then for any $hk \in HK$ with $h \in H, k \in K$ and any $g \in G$, we get

$$ghkg^{-1} = \underbrace{(ghg^{-1})}_{\in H} \underbrace{(gkg^{-1})}_{\in K} \in HK,$$

so HK is closed under conjugation, hence normal. □

Problem 8. The **commutator subgroup** of a group G is the group G' generated by the elements $[x, y] = x^{-1}y^{-1}xy$, where x and y vary through the elements of G .

- (a) Show that G' is normal in G .
- (b) Show that G/G' is an abelian group.
- (c) Let H be a subgroup of G containing G' . Show that H is normal in G , and that G/H is abelian.
- (d) Suppose that N is a normal subgroup of G such that G/N is abelian. Show that $G' \subseteq N$.

Solution. (a) First, observe that the conjugate $h^{-1}[x, y]h$ of a commutator is also a commutator $[h^{-1}xh, h^{-1}yh]$. Then, an element g of G' is a product of finitely many commutators, say

$$g = [x_1, y_1][x_2, y_2] \cdots [x_n, y_n].$$

For any $h \in G$,

$$h^{-1}gh = (h^{-1}[x_1, y_1]h)(h^{-1}[x_2, y_2]h) \cdots (h^{-1}[x_n, y_n]h) = \prod_{i=1}^n [h^{-1}x_ih, h^{-1}y_ih]$$

is thus a product of commutators, so G' is closed under conjugation, hence is normal.

(b) Let $\varphi: G \rightarrow G/G'$ be the projection homomorphism, i.e. $\varphi(x)$ gets sent to the coset of G' containing x ; this is a surjective homomorphism with kernel G' . Then, given $g, h \in G/G'$, let x and y be elements of the cosets g, h ; we have

$$gh = \varphi(x)\varphi(y) = \varphi(xy) = \varphi([x^{-1}, y^{-1}]yx) = \varphi([x^{-1}, y^{-1}])\varphi(xy) = \varphi(xy) = hg$$

because the commutator $[x^{-1}, y^{-1}]$ is in G' , so G/G' is abelian.

(c) If $h \in H$, then $h^{-1} \in H$ as well. Thus for any $g \in G$,

$$g^{-1}hg \in H \iff h^{-1}(g^{-1}hg) = [h, g] \in H,$$

but we know that $[h, g] \in H$ because H contains all the commutators in G . Thus, H is closed under conjugation, hence normal.

It is straightforward to show that G/H is abelian, with the exact same logic as part (b). Let $\varphi: G \rightarrow G/H$ be the projection homomorphism. Then, given $g, h \in G/H$, let x and y be elements of the cosets g, h ; we have

$$gh = \varphi(x)\varphi(y) = \varphi(xy) = \varphi([x^{-1}, y^{-1}]yx) = \varphi([x^{-1}, y^{-1}])\varphi(xy) = \varphi(xy) = hg$$

because the commutator $[x^{-1}, y^{-1}]$ is in $G' \subseteq H$, so G/H is abelian.

(d) If G/N is abelian, let $\varphi: G \rightarrow G/N$ be the projection homomorphism. Then, for any $x, y \in G$, we have

$$\varphi(x)\varphi(y) = \varphi(y)\varphi(x) \iff \varphi(x^{-1}y^{-1}xy) = 1,$$

so every commutator is in N , hence the subgroup generated by the commutators, G' , is contained in N . $\square$